6.1 Dilations

Lesson Introduction

<table>
<tr><td>Benchmark</td><td colspan="2">MA.912.GR.2.1 Given a preimage and image, describe the transformation and represent the transformation algebraically using coordinates.</td><td>Learning Targets</td><td colspan="2">• I can write an algebraic description using coordinate notation of a dilation on a coordinate plane.
• I can describe a dilation by giving the scale factor and center of dilation.</td></tr>
<tr><td rowspan="2">Benchmark Coherence</td><td colspan="3">Building On</td><td colspan="2">Working Toward</td></tr>
<tr><td colspan="3">MA.8.GR.2.2 Given a preimage and image generated by a single dilation, identify the scale factor that describes the relationship.</td><td colspan="2">N/A</td></tr>
<tr><td>Essential Questions</td><td colspan="5">• What is a scale factor?
• What is a center of dilation?
• How do you write the algebraic description of a dilation?</td></tr>
<tr><td>Lesson Narrative</td><td colspan="5">In this lesson, the students will be determining the scale factor of a dilation and writing an algebraic description using the scale factor. Connections will be made between using the points of the preimage and image to find the scale factor, and using the slope and length of sides of preimages and images.</td></tr>
<tr><td rowspan="3">Component Summary</td><td>6.1.1 Warm-Up (~5 min.)</td><td>Drawing using point of perspective (SE p. 285)</td><td>6.1.4 Guided Instruction (10-15 min.)</td><td colspan="2">Writing the algebraic description of a dilation (SE p. 288)</td></tr>
<tr><td>6.1.2 Refresher (5 min.)</td><td>Features of a dilation (SE p. 286)</td><td>6.1.5 Guided Practice (5-10 min.)</td><td colspan="2">Practice writing the algebraic description of a dilation (SE p. 289)</td></tr>
<tr><td>6.1.3 Collaboration (20-25 min.)</td><td>Determining the scale factor of a dilation in multiple ways (SE p. 286)</td><td>6.1.6 Wrap-Up (~5 min.)</td><td colspan="2">Students summarize their learning (SE p. 290)</td></tr>
<tr><td rowspan="2">Resources</td><td>Glossary</td><td colspan="2">Materials</td><td colspan="2">Additional Preparation</td></tr>
<tr><td><u>Key Terms:</u> N/A

<u>Additional Terms:</u>
• Center of Dilation
• Scale Factor
• Preimage
• Image</td><td colspan="2">• (Optional) Straightedge</td><td colspan="2">• Throughout this lesson, students will be exposed to multiple methods of determining the scale factor of a dilation. Make sure, as the teacher, you are proficient in each method so you can support students with each method.</td></tr>
</table>

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may not know the difference between the image and the preimage.
Students may incorrectly state the scale factor by swapping the numerator and the denominator of the ratio.
Students may not correctly determine the slope of lines when using a different scale on the x-axis from the y-axis.</td><td>Consider providing a quick refresher (no more than a couple of minutes) on determining the length of a line segment using the distance formula.
Determine if your students need the support of all or part of the 6.1.2 Refresher.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Anticipate, Monitor, Select, Sequence, Connect
In 6.1.3 Collaboration, students are asked to make connections between determining the scale factor from points on the preimage and image to finding the scale factor from the slope and from the length of a line segment. Implement the monitor and select part of this routine by selecting students for whole-group discussion that will result in students hearing the connections made from multiple sources.</td><td>MA.K12.MTR.2.1: Multiple Representations
Throughout the lesson, students are exposed to multiple methods of determining the value of the scale factor for a given dilation. Students are given opportunities to express connections between the methods so that they can make decisions on which method to choose given the context of the problem.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">When determining student pairings for the 6.1.3 Collaboration, consider pairing confident students with students who need additional support. Set the expectation that each student will contribute and guide the other in making connections.

Students are first introduced to dilations and scale factor in Grade 8 Unit 9. Consider using this unit as a resource and additional entry-point for reviewing the content to prepare students for the content of the lesson.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

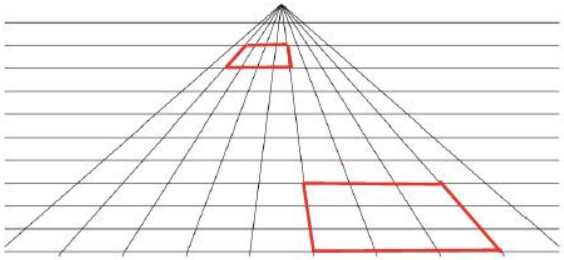
6.1.1 Warm-Up: What Is Your Perspective?

Component Overview: *Students observe how parallel lines in a three-dimensional drawing are represented using point perspective.*

6.1.1 Warm-Up: What Is Your Perspective?

Point perspective is a common method of drawing a three-dimensional object onto a two-dimensional plane. The artist chooses one point from which everything in the drawing is set. It was first discovered during the Renaissance. An example of a painting that uses point perspective is Vincent van Gogh's *Vincent's Bedroom in Arles*.

1. Use the grid to sketch a rectangle that is close to the point of perspective and a rectangle that is farther away from the point of perspective.



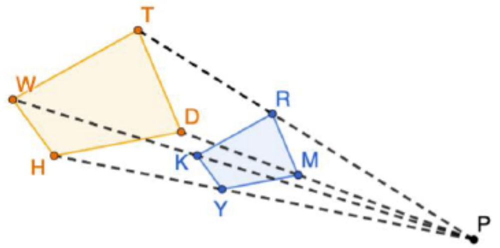
The one point the artist chose to draw everything from is considered the "point of dilation." This question allows for discussion of how point perspective is connected to dilations. If time permits, return to 6.1.1 Warm-Up at the end of the lesson to probe students about this connection.

6.1.2 Refresher: Dilations

Component Overview: *Students are given a word bank to scaffold their current knowledge of dilations and prepare them to apply the algebraic description of dilations. Take time, through discussion of the answers, to assess if students have a solid grasp of this prerequisite knowledge before moving into the lesson content.*

6.1.2 Refresher: Dilations

1. $KRMY$ was created by dilating $WTDH$. The ratio of WT to KR is $\frac{5}{3}$.



a. Use the words in the bank below to complete the statements that follow.

center of dilation	image
pre-image	scale factor

Point P is the __*center of dilation*___.

$KRMY$ is the _*image*_______________.

$WTDH$ is the __*pre-image*_____________.

The ____*scale factor*_____ is $\frac{3}{5}$.

b. Use the diagram to determine each.

What is the ratio of $\frac{WH}{KY}$? $\frac{5}{3}$

What is the ratio of $\frac{PR}{RT}$? $\frac{3}{2}$

Consider asking the following:

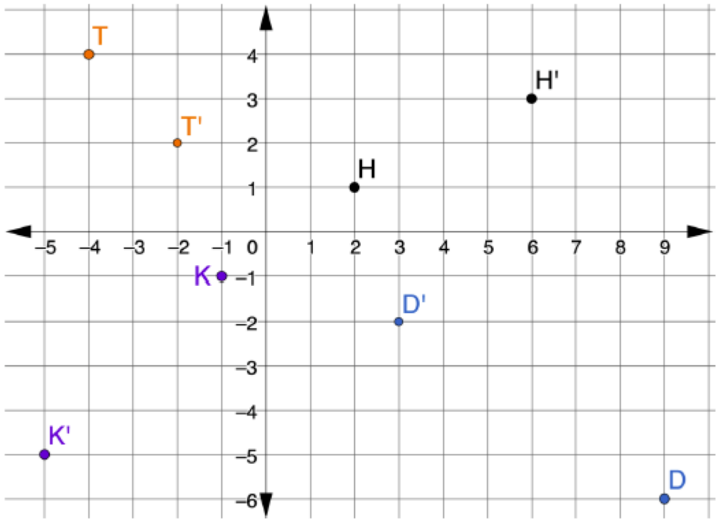
- *Why is it important to know where your center of dilation is?*
- *What is the difference between the image and preimage?*
- *What is a scale factor?*

6.1.3 Collaboration: Algebraic Description of a Dilation

Component Overview: *Students are introduced to the concept of dilations in the coordinate plane. Students will examine the positions of preimage points and image points to determine the scale factor and write an algebraic representation of the dilation.*

6.1.3 Collaboration: Algebraic Description of a Dilation

1. Points D, H, K, and T are shown on the coordinate grid. Each point has been dilated using $(0,0)$ as the center of dilation.



a. Complete the table.

Point	Ordered Pair	Ordered Pair of the Dilation	Scale Factor
D	$(9,-6)$	$(3,-2)$	$\frac{1}{3}$
H	$(2,1)$	$(6,3)$	3
K	$(-1,-1)$	$(-5,-5)$	5
T	$(-4,4)$	$(-2,2)$	$\frac{1}{2}$

In the coordinate plane, the algebraic representation of a dilation centered at the origin is $(x,y) \rightarrow (kx,ky)$, where k is the scale factor.

Begin this component by providing students an opportunity to make observations first, prompting them to make connections between the image and preimage.

- *What do you notice about the ordered pair and the ordered pair of the dilation?*
- *How does this relationship lead to finding the scale factor?*

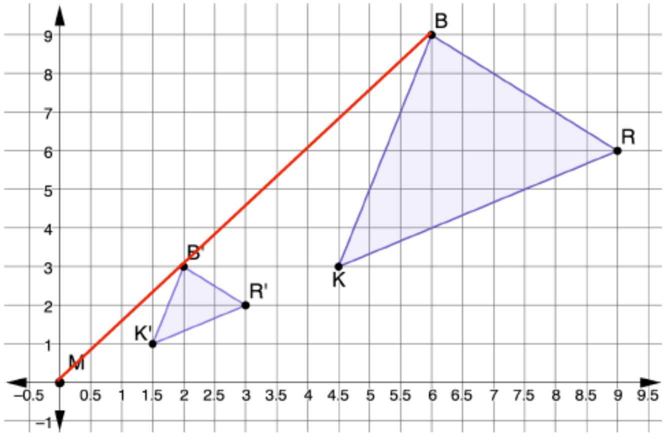
Each of these points are a different representation of a dilation. They are not connected to each other. Make sure to spend time addressing how to write the algebraic representation for each set of points.

6.1.3 Collaboration: Algebraic Description of a Dilation (continued)

b. Write the algebraic representation for each point.

Point	Algebraic Representation of Dilation
D	$(x, y) \rightarrow (\frac{1}{3}x, \frac{1}{3}y)$
H	$(x, y) \rightarrow (3x, 3y)$
K	$(x, y) \rightarrow (5x, 5y)$
T	$(x, y) \rightarrow (\frac{1}{2}x, \frac{1}{2}y)$

2. $\Delta R'B'K'$ is the result of dilating ΔRBK using the origin as the center of dilation.



a. What is the scale factor of the dilation?
The scale factor is $\frac{1}{3}$

b. Write the algebraic description of the dilation.
$(x, y) \rightarrow \left(\frac{1}{3}x, \frac{1}{3}y\right)$

c. Draw a line segment from point M that passes through B' and ends at B.
See drawing.

Each of these points represents the same dilation of the given figure. While monitoring, record which points students used to determine the scale factor. Call on these students to demonstrate how they determined the scale factor, making the connection that any corresponding points in the preimage and image can be used.

6.1.3 Collaboration: Algebraic Description of a Dilation (continued)

d. The table shows different relationships of the line segments MB, $B'B$, and MB'.

Line Segment	Slope	Distance
MB'	$\frac{3}{2}$	$\sqrt{2^2+3^2}=\sqrt{13}$
$B'B$	$\frac{6}{4}$	$\sqrt{6^2+4^2}=\sqrt{52}=2\sqrt{13}$
MB	$\frac{9}{6}$	$\sqrt{6^2+9^2}=\sqrt{117}=3\sqrt{13}$

e. Discuss with your partner how the scale factor could be determined using slope. Summarize your discussion.
Answers will vary. The scale factor can be found using the ratio of the rise for the slope of MB' to the rise of the slope of MB.

f. Discuss with your partner how the scale factor could be determined using distance.
Answers will vary. To find the scale factor using distance write the ratio of "the distance from the point of dilation to the image point" to "the distance from the point of dilation to the corresponding pre-image point". For example, using Point B' to Point B, this ratio is $\frac{\sqrt{13}}{\sqrt{117}}=\frac{\sqrt{1}}{\sqrt{9}}=\frac{1}{3}$.

Some students may understand a scale factor using the slope or distance. This may be helpful when students are drawing a dilation.

- *Why is using the slope or distance an acceptable method to use to find dilations?*
- *How can you use the slope or distance for finding dilations?*

Students do not need to write a response for question 2f. It is a discussion opportunity. A sample response is included for reference as teachers listen to student discussions.

6.1.3 Collaboration: Algebraic Description of a Dilation (continued)

3. The table shows different relationships among the line segments BR, $B'R'$, KR and $K'R'$.

Line Segment	Slope	Distance
BR	$-\frac{3}{3}$	$\sqrt{3^2+3^2}=\sqrt{18}=3\sqrt{2}$
$B'R'$	$-\frac{1}{1}$	$\sqrt{1^2+1^2}=\sqrt{2}$
KR	$\frac{3}{4.5}$	$\sqrt{3^2+4.5^2}=\sqrt{29.25}$
$K'R'$	$\frac{1}{1.5}$	$\sqrt{1^2+1.5^2}=\sqrt{3.25}$

Discuss with your partner how the scale factor is also found in the relationships between the slopes of BR and $B'R'$, and KR and $K'R'$. Write a summary of the relationships.

When writing the ratio of the numerators (rise) of the slopes in the image to the numerators (rise) of the slopes of the pre-image, you get a ratio of $\frac{1}{3}$. You also get the same ratio when writing the ratio of the denominators (runs). The ratio of the length of each line segment of the image to the length of the corresponding line segment of the pre-image is equal to the scale factor, $\frac{1}{3}$. For example, $\frac{B'R'}{BR}=\frac{\sqrt{2}}{3\sqrt{2}}=\frac{1}{3}$.

In question 2, the students examine the lines drawn from the point of dilation to corresponding points on the preimage and image. In question 3, the students examine the lines between points on a preimage and compare those to the lines between corresponding points on the image. Probe students to make sure they see the similarities and differences between these two questions.

Consider providing graph paper to provide a visual entry-point with question 3.

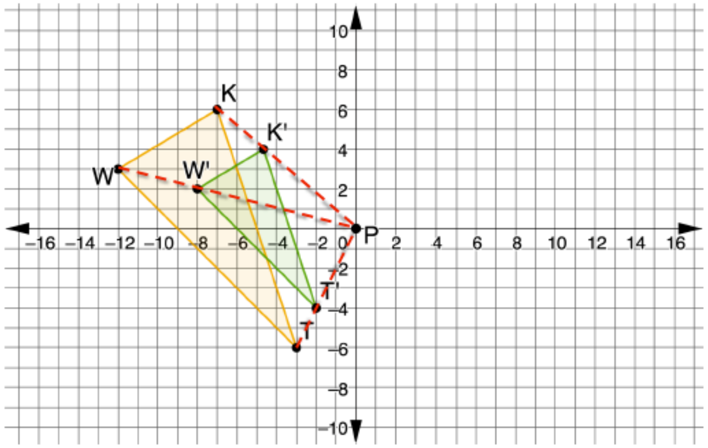
6.1.4 Guided Instruction: Describing Dilations

***Component Overview**: Students are guided through the multiple methods of determining the scale factor of a dilation. Exposing students to different methods equips them to make a choice between methods depending on how dilation information is provided.*

6.1.4 Guided Instruction: Describing Dilations

1. $\Delta K'M'T'$ is the result of dilating ΔKMT using the origin as the center of dilation. The vertices of ΔKMT are located at $K(-8,-6)$, $M(6,7)$, and $T(0,4)$. The vertices of $\Delta K'M'T'$ are located at $K'(-12,-9)$, $M'(9,10.5)$, and $T'(0,6)$. Write the algebraic description of the dilation.
   $(x,y) \rightarrow (\frac{3}{2}x, \frac{3}{2}y)$

2. Triangles TWK and $T'W'K'$ are shown on the coordinate grid, where $\Delta T'W'K'$ is a dilation of ΔTWK and point P is the center of dilation.



   a. Show the dilation on the coordinate grid by drawing dotted lines from the center of dilation through each vertex.
   See image.

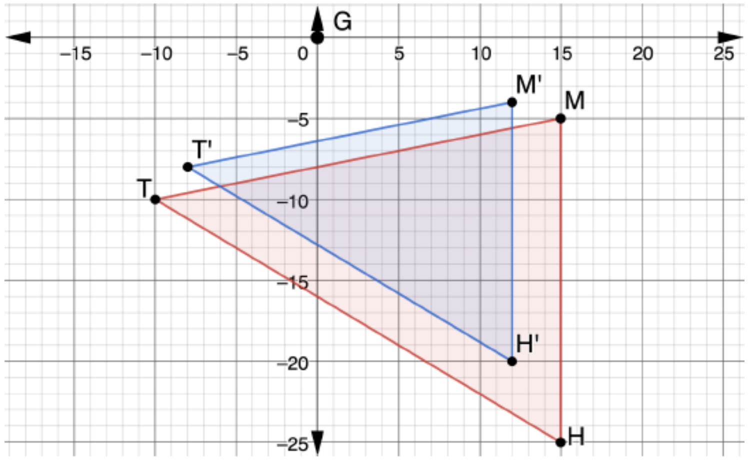
   b. What is the scale factor of the dilation?
   The scale factor is $\frac{2}{3}$.

   c. Write the algebraic description of the dilation.
   $(x,y) \rightarrow (\frac{2}{3}x, \frac{2}{3}y)$

Students may prefer one method of determining scale factor over another. For each of these questions, expose students to all methods for them to make decisions based on what they understand and how the information is given.

6.1.4 Guided Instruction: Describing Dilations (continued)

3. Triangles THM and $T'H'M'$ are shown on the coordinate grid, where $\Delta T'H'M''$ is a dilation of ΔTHM.



Write a description of the dilation by giving the center of dilation and the scale factor.

Answers will vary. A dilation of ΔTHM centered at point G (or the origin) using a scale factor of $\frac{4}{5}$ created $\Delta T'H'M'$.

Students can use a variety of ways to determine the scale factor. For students who are struggling to determine coordinates, consider suggesting that the students use GT' and GT. Equipping students with multiple methods of finding a dilation help students flexibly solve a variety of problems.

Explain how the collinearity of corresponding vertices shows the orientation of the shapes is the same. Explain that when the orientation of the shapes is not the same, but the shapes are similar, a transformation was applied before or after a dilation.

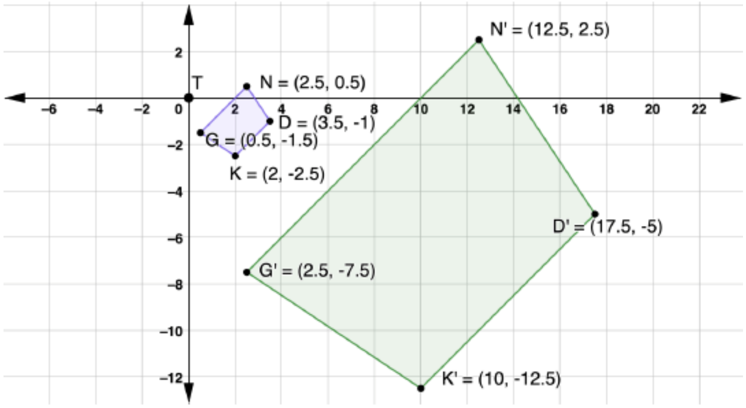
6.1.5 Guided Practice: Describing Dilations

Component Overview: *Students will describe dilations and write the algebraic representation of the dilation. Allow students to choose the method they prefer for determining the scale factor.*

6.1.5 Guided Practice: Describing Dilations

1. $\overline{M'T'}$ has coordinates $M'(-16, 20)$ and $T'(4, -8)$, and is the result of the dilation of $\overline{MT}$ centered at the origin. The coordinates of $\overline{MT}$ are $M(-4, 5)$ and $T(1, -2)$. Write the algebraic description of the dilation.
   $(x, y) \rightarrow (4x, 4y)$

2. $NDKG$ and $N'D'K'G'$ are shown on the coordinate grid, where $N'D'K'G'$ is a dilation of $NDKG$.



   Write a description of the dilation by giving the center of dilation and the scale factor.
   Answers will vary. The quadrilateral $NDKG$ centered at point T (or the origin) using a scale factor of 5 created the quadrilateral $N'D'K'G'$.

While students are working, monitor and make note of which methods students are using. When debriefing these questions, select students who used different methods to demonstrate their work.

To add a challenge, have students demonstrate how to find the scale factor using another method than the one they originally chose. This move will provide students with a way of analyzing the accuracy of their answers.

6.1.6 Wrap-Up: Cool Down

***Component Overview**: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.*

6.1.6 Wrap-Up: Cool Down

1. Explain how dilations are different from rigid transformations, in your own words.
   Answers will vary.

The information provided from the students here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.